



Learning real-world heterogeneous noise models with a benchmark dataset

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ABSTRACT

Noise modeling is an important research field in computer vision; the design of an accurate model for imaging sensor noise depends on not only a comprehensive benchmark dataset of the real world, but also a precise design of the noise modeling algorithm. However, due to the inaccurate estimation method of noise-free images and limited shooting scenes, the current realistic datasets could not describe the diverse noise properties sufficiently. Moreover, popular parametric noise models are not sophisticated enough to characterize the real-world noise exactly. In this work, we first construct a more comprehensive dataset of the real world by capturing more indoor and outdoor scenes under different lighting conditions using diverse smartphones, then we propose a non-parametric noise estimation method capable of modeling the spatial heterogeneity of real-world noise patterns. Specifically, in order to characterize the spatial heterogeneity of real-world noise, we assume a non-i.i.d Gaussian distribution and propose a deep convolutional neural network (DCNN)-based approach for learning pixel-wise noise variance maps. To learn the pixel-wise variance map, we have constructed a variance estimation network mapping from the conditional signals (clean image, ISO, and camera model) to surrogate labels obtained from the nonlocal search of similar patches from the clean-noisy image pair. Finally, we conducted denoising and classification experiments using different kinds of simulated noisy images, compared to the Poisson-Gaussian and Noise Flow noise models, the proposed method achieves denoising performance improvements (PSNR) of 1.13 dB and 2.51 dB respectively on the proposed real-world test dataset, denoising and classification results on the real noisy data captured by mobile phones have verified that our approach is more accurate than current noise modeling methods.

1. Introduction

Due to the difficulty of obtaining clean-noisy image pairs in the real-world [1], current DCNN-based image denoising algorithms [2–4] have focused on the removal of simulated noise, such as additive white Gaussian noise (AWGN). These methods could achieve state-of-the-art performance on the simulated noisy datasets, but their performance often degrades rapidly on the real noisy one. Some DCNN-based denoising methods even perform worse than traditional model-based approaches such as [5,6], which has been known as “simulated-to-real gap” [7]. Filling in such a gap requires accurate modeling of the noise distribution in the real-world scenario.

Noise modeling is the process of characterizing the various sources of noise present in data acquisition and measurement systems, which involves understanding the statistical properties of noise, such as its distribution, amplitude and spatio-temporal correlations, while denoising aims at removing noise from data while preserving the underlying clean signal. Establishing an accurate noise model is significant for

the following two reasons. Firstly, it can distinguish clean signal and noise, enabling more accurate interpretation and analysis. In addition, accurate noise modeling is essential for designing effective denoising algorithms, e.g., more realistic noisy data can be generated to prompt the real-world denoising performance of data-driven deep learning methods.

For exactly modeling the noise distribution of the real world, a comprehensive and accurate realistic benchmark dataset is necessary in the first place. Some literature [1,8–10] have built corresponding realistic clean-noisy image datasets by capturing short-exposure and long-exposure images. However, there still exist some limitations to these methods. Firstly, some methods [1,8] employ long-exposure images as noise-free images. In this way, the quality of reference images would rely on the careful design of post-processing operations because the noise in long-exposure and low-ISO images may be still noticeable. Moreover, other algorithms [9,10] employ a time-consuming but more plausible method that shoots the same scene many times and takes a

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weighted average of these images to derive noise-free images, e.g., the PloyU dataset [9] and the SIDD dataset [10]. However, the PloyU dataset [9] contains 40 different scenes and derives 40 noisy-clean image pairs which are limited and cannot sufficiently describe diverse noise properties, and the SIDD dataset [10] has not considered outdoor scenarios and corresponding noise characteristics. Therefore, a more comprehensive benchmark dataset of the real world is in desperate need.

Regarding the task of real-world noise modeling, it is often complicated by the composition of diverse types of noise introduced by different imaging stages [11] such as photon shot noise, read noise, banding pattern noise, quantization noise, and so on [12]. A principled approach toward noise modeling has to take as many uncertainty factors (both camera-related and environment-related) as possible into account. The most commonly used noise models nowadays are the parametric ones that can be further classified into homoscedastic (e.g., additive Gaussian noise [2–4]) and heteroscedastic (e.g., Poisson-Gaussian noise [13]). However, parametric noise modeling methods have several limitations. First, one has to estimate the hyperparameter from limited observation data — there exists an intrinsic trade-off between the model complexity (e.g., the number of hyperparameters [14]) and the data characteristics (e.g., paired or unpaired [15]). To make the problem more tractable, one has to make over-simplified assumptions, such as ignoring the signal-dependent characteristics of i.i.d. Gaussian noise models [5] and overlooking heterogeneity in Poisson-Gaussian noise models [13].

Unlike parametric noise models, data-driven approaches have shown a significantly better capability of characterizing the complexity of real-world noise [16]. These recent noise models can be classified into two categories: variational inference network (VIN)-based methods [17–20] and GAN-based methods [21–23]. The former tackles the problem of blind noise estimation by combining the transparency of well-established parametric models with the flexibility and expressiveness of deep neural networks (e.g., normalizing flow network [18,19] and sigma network [17]). The latter leverages the idea of adversarial learning [24] to generate synthetic noise that is indistinguishable from the real one. However, these methods still suffer from various weaknesses, such as the over-simplification of noise models and the over-dependency on hyperparameters. Furthermore, it is known that the performance of the VIN-based methods requires careful incorporation of prior knowledge (e.g., ISO numbers and camera models) [17]; and the training process of GAN often faces the issue of stability [25].

In this paper, we first establish a more comprehensive noisy-clean image dataset of the real world to better characterize realistic noise properties. Due to the prevalence of mobile photography, we focus on modeling imaging sensor noise (i.e., RAW format images) introduced by smartphone photography. Moreover, we propose a principled solution for modeling real-world noise. We assume that real-world noise observes non-i.i.d Gaussian distribution due to the heterogeneous properties of noise sources. Accordingly, we propose to estimate the noise variance on a pixel-by-pixel basis, which can characterize the noise variance at the finest scale. Toward this objective, we have constructed a DCNN-based variance estimation network to learn the mapping from conditional inputs (clean image, ISO, and camera model) to a desirable per-pixel variance map. To verify the accuracy of the proposed noise modeling method, we have trained several image denoising and classification networks using different kinds of artificial noisy data and evaluated the performance on realistic noisy test sets. Experimental results have verified the accuracy of the proposed noise modeling method. Our main contributions are summarized as follows.

- Real-world benchmark dataset for real denoising task. By considering more indoor and outdoor scenarios under different lighting conditions and estimating noise-free images using a principled pipeline, we have built a more comprehensive realistic dataset.

- Pixel-wise variance estimation network. We have developed a nonlocal block matching-based method for obtaining surrogate variance maps from paired clean/noisy images. Such surrogate labels will serve as the desirable output during the training of DCNN-based variance map estimation.
- Improved real-world denoising and classification performance. Experimental results on realistic noisy and classification test sets confirm that the proposed method outperforms other computing methods in terms of both PSNR value and classification accuracy.

The remainder of this paper is organized as follows. Previous researches related to our work are reviewed in Section 2. To describe comprehensive noise characteristics, a real-world benchmark dataset is developed in Section 3. Next, we describe the proposed noise modeling method and the details of each component in Section 4. Subsequently, the experimental settings, comparative results and analyses are presented in Section 5. Finally, we conclude the paper and discuss the limitation and future work.

2. Related work

2.1. Real-world clean-noisy image datasets

According to the methods that generate noise-free images, current datasets could be classified into two categories. On the one hand, some methods [1,8] capture short-exposure and long-exposure images respectively under the same scene, and they employ the low-ISO and long-exposure images as the ground truth images. For example, the RENOIR dataset [8] is the first real-world image dataset that contains realistic noisy and clean image pairs. However, the insufficient post-processing operations limit the quality of the proposed dataset. In particular, the RENOIR dataset [8] has not conducted spatial alignment, and the noise in low-ISO images is still perceptible, thus the proposed ground-truth images are not accurate. Furthermore, the DND dataset [1] conducts post-processing operations to the original captured low-ISO images and uses the resulted images as clean images. The alignment and other post-processings improve the quality of reference images, yet the DND dataset [1] only concentrates on normal lighting conditions and ignores low-light scenes.

On the other hand, other methods [9,10] choose to generate reference images in a more complicated but reasonable manner. Specifically, they shoot the same scene many times and obtain the corresponding images, then they apply image post-processing operations (e.g., image alignment and abnormal image removal) to these captured images. Finally, the clean image could be generated by weighted averaging these resulted images. [9] proposed the PloyU dataset, while the corresponding noise-free images are obtained by capturing the same scene for 500 times and averaging these noisy photographs. Due to the time-consuming shooting, the PloyU dataset only contains 40 clean-noisy image pairs, which are insufficient for characterizing diverse noise properties. In [10], the authors propose a complete pipeline to estimate clean images from redundant noisy images. The proposed SIDD dataset [10] has been a large realistic benchmark dataset in recent years. However, the SIDD dataset only captured the indoor scenes and has not considered the noise characteristic of outdoor scenes.

2.2. Parametric noise modeling methods

The most common noise model in image denoising literature is the so-called additive white Gaussian noise (AWGN): $n_i \sim \mathcal{N}(0, \sigma^2)$, where n_i denotes the intensity value of noise at a position i , and σ denotes the standard deviation of noise. Despite its simplicity, AWGN has limited modeling capability and cannot characterize sensor noise that is often signal-dependent (e.g., photon shot noise) [12]. To overcome the limitation of AWGN, the Poisson-Gaussian noise model [13] has

been developed, in which signal-dependent property can be characterized by a new noise level function (NLF) [12]. More specifically, the Poisson-Gaussian noise model consists of both signal-dependent and signal-independent components [13] - i.e.,

$$n_i \sim \mathcal{N}(0, \sigma_r^2 + \sigma_s^2 x_i), \quad (1)$$

where x_i is the clean pixel at position i , and the value of σ_r , σ_s are jointly determined by sensor parameters and environmental factors such as the analog gain, the ISO of cameras, and the ambient temperature [13]. More recently, the Gaussian mixture model (GMM) [26] was proposed as a more generic and flexible solution to approximate arbitrary continuous distribution. However, its accuracy and optimality for modeling real-world noise are still questionable [26]. More accurate learning-based noise models [11,27] have been developed in recent years, which attempt to consider the physics of imaging sensors and image processing pipeline (e.g., gain, color correction, and tone mapping) [27]. However, the hyperparameters in those data-driven models need to be calibrated for every camera device and different imaging conditions, which becomes cumbersome.

2.3. Non-parametric noise modeling methods

Historically, nonparametric approaches toward noise modeling have received less attention, with few exceptions [28]. Only in recent years, non-parametric noise modeling methods are developed upon exploiting the powerful representation capability of deep neural networks for learning the complicated distribution of real-world noise. As of today, deep learning-based noise models can be mainly classified into two categories: VIN-based methods and GAN-based methods. Variational auto-encoder [29] belongs to unsupervised learning to aim at transforming one distribution to another. In [17] a variational inference method was proposed to infer the distribution of the underlying clean image and noise distribution directly from the observed noise image under a unified Bayesian framework. Using the basic architecture of normalizing flow [30], Noise Flow was introduced in [18] which consists of a series of bijective functions. The network parameters of Noise Flow were updated by minimizing the loss function derived from a maximum likelihood estimation. In [19], an improved structure was proposed to jointly train the noise model and denoiser that does not require clean data.

By contrast, GAN [24] has found successful applications in simulating complex and realistic distribution. For example, GAN-CNN-based blind denoiser (GCBDD) [31] used a GAN architecture to learn the mapping from noisy images to real noise. Grouped residual dense network (GRDN) [22] proposed a more realistic noise modeling method that exploits the conditional signals (e.g., clean images, ISO, and camera information) to learn the noise distribution. In view of the difficulty with paired clean/noisy images, self-consistent GAN (SCGAN) [21] proposed an unsupervised GAN-based noise modeling method for unpaired training data. Most recently, dual-adversarial network (DANet) [23] proposed to estimate two different factorization forms of the joint distribution and led to the joint training of the denoising network and noise generation network. The camera-aware noise modeling method was developed in [16] by combining a camera-encoding network with a noise-generating network.

2.4. Classification, segmentation and object detection methods

Classification, segmentation and object detection are fundamental tasks in the field of high-level computer vision. Over the decades, significant advancements have been made in image classification, segmentation and object detection largely driven by deep learning techniques.

Image classification is the process of categorizing digital images into different classes or labels, classical CNN architectures, such as ResNet [32], DenseNet [33] and MobileNet [34], have achieved remarkable performance on benchmark datasets like ImageNet [35],

Table 1

Different smartphones and shooting settings.

Smartphone	ISO	# of images
Huawei P20	100–1600	15
Redmi K40	100–6400	36
Redmi K30	100–3200	10
Xiaomi Mi 10	100–3200	37
Apple iPhone 7 Plus	100–1600	34
VIVO X30 Pro	100–3200	8

useful techniques like residual connections, densely connected layers and efficient network scaling are employed to enhance classification accuracy while maintaining computational efficiency.

Image segmentation involves partitioning an image into multiple segments or regions based on certain criteria, such as color, intensity or texture similarity. Unlike classification, segmentation provides pixel-level information about object boundaries, mainly including semantic segmentation and instance segmentation approaches, where U-Net [36], SegNet [37] and Mask R-CNN [38], YOLACT [39] are classical architectures of semantic segmentation and instance segmentation respectively. Most recently, [40] presented a software application that can be easily used by users for automatic parotid gland tumor (PGT) segmentation.

Object detection aims to identify specific objects and accurately localize them. DCNNs enable the creation of highly accurate and efficient models, such as one-stage detector, e.g., YOLO [41] and two-stage detector, e.g., Faster R-CNN [42]. Further research has focused on improving the speed, accuracy and robustness of detection models, techniques such as feature pyramid networks (FPN) [43] and anchor-free [44] methods have been proposed to enhance detection performance, particularly for small or overlapping objects.

3. Proposed real-world benchmark dataset

The SIDD dataset [10] has been a widely-used benchmark dataset recently. However, this realistic dataset only captured indoor scenes, and the number of scenes is only 10 which is limited. To tackle this problem, we consider outdoor and indoor scenes under different lighting conditions, we also expand the number of scenes to better characterize diverse noise properties, which means we have built a more comprehensive benchmark dataset of the real world. To generate high-quality noise-free images, we capture the same scene many times and then generate the corresponding reference image by the similar ground truth estimation pipeline in [10].

The types of smartphones, the ISO settings and the number of image pairs are demonstrated in Table 1. Specifically, we use 6 different common types of smartphones (e.g., Huawei P20, Redmi K40, Redmi K30, Xiaomi Mi 10, Apple iPhone 7 Plus and VIVO X30 Pro) to capture realistic noisy images in the raw domain, the values of ISO range from 100 to 6400, and the numbers of image pairs of each smartphone from 8 to 37. To continuously capture noisy data of the same scene, the smartphones are fixed to a tripod to keep the images well-aligned during shooting. To explore the influence of different lighting conditions on noise characteristics, we capture normal-light and low-light images for indoor scenes, and normal-light images for outdoor scenes. Note that the processing pipelines of indoor scenes and outdoor scenes are slightly different due to environment-based reasons. When capturing noisy images of indoor scenes, we continuously shoot 180 images for each scene and then use the ground truth estimation pipeline of [10] to derive the estimated reference image. Regarding outdoor scenes, we only capture continuous 30 images to avoid the object motions brought by uncontrollable reasons (e.g., wind blowing). Since the number of shooting images is small under outdoor scenes, some noise may persist in the derived reference image. Therefore, we use an image denoising method (e.g., BM3D [5]) for further noise removal to get the ultimate noise-free image.



Fig. 1. Visualizations of some image patches sampled from ground-truth images (left) and noisy images (right) of the proposed dataset. These patches of raw-data passed through the ISP pipeline provided by SIDD [10] for visualization (Please zoom in to see better details).

Table 2

The quantitative comparison of the SIDD dataset and the developed benchmark dataset.

Datasets	The SIDD dataset	The developed dataset
The number of indoor scenes	10	12
The number of outdoor scenes	0	7
The number of cameras	5	6

In total, the proposed dataset contains 140 image pairs which include 101 indoor image pairs and 39 outdoor image pairs. To further enrich the diversity of scenes, we expand the number of scenes to 19, with 12 indoor scenes and 7 outdoor scenes. As demonstrated in Table 2, the proposed dataset has considered more outdoor scenes and explored more diverse noise characteristics compared to the SIDD dataset [10], and some patches sampled from the noise-free images and noisy images of the constructed dataset are as shown in Fig. 1.

4. Proposed noise modeling method

In order to characterize the heterogeneity of the real-world noise, we advocate an approach of estimating the noise variance on a pixel-by-pixel basis. Such per-pixel variance estimation allows us to model the heterogeneity at the finest scale. In view of the dramatically increased number of hyperparameters, we propose to construct a devoted variance estimation network. Unlike [21], we assume access to conditional inputs (e.g., clean image, ISO number and camera model) to optimize the modeling accuracy at the pixel instead of patch or image level.

In this section, we first present a pixel-wise variance estimation method based on the non-local block matching of clean-noisy image pairs, which will serve as the proxy label for the proposed DCNN-based noise modeling. We have verified the accuracy of this non-i.i.d Gaussian noise assumption based on the pre-computed pixel-wise variance map. To synthesize a more realistic noise distribution, we propose a pixel-wise variance estimation network based on the analogy between image restoration and noise estimation. Specifically, considering the characteristic of noise is related to conditional signals (e.g., clean image, camera models, ISO and other imaging-related information), we have leveraged the DCNN architecture to learn the mapping function from conditional signals to the pixel-wise variance map. The overall architecture is shown in Fig. 3.

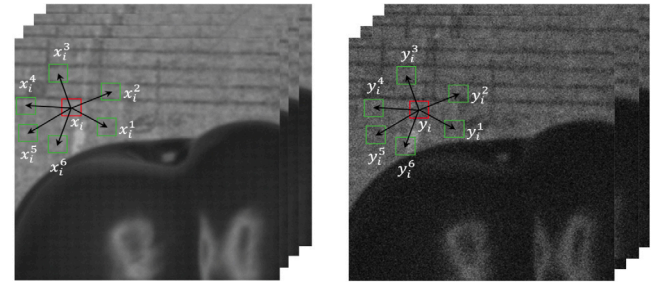


Fig. 2. Generating per-pixel variance map from clean-noisy image pair by non-local search of similar patches [6].

4.1. Per-pixel variance map estimation

Given real-world noisy image data, we consider the following additive observation model [5,45]

$$\mathbf{y} = \mathbf{x} + \mathbf{n}, \quad (2)$$

where $\mathbf{x}$, $\mathbf{y}$ denote the clean and noisy images respectively, and $\mathbf{n}$ is the additive noise term. Image restoration and noise estimation are two sides of the same coin. In the scenario of simulated noise (known noise statistics), we focus on the estimation of signal variance σ_x because it helps the derivation of the optimal filter (e.g., Wiener filtering). In this work, our focus shifts to the estimation of noise variance σ_n and it follows that clean image $\mathbf{x}$ becomes the interference. To facilitate the estimation of per-pixel noise variance maps, we use clean-noisy image pairs of the proposed dataset as training data.

Inspired by the analogy between image restoration and noise estimation, we propose to compute the variance map of each pixel by leveraging the existing nonlocal block matching method for finding similar patches (e.g., [6]). Assuming i denotes the position index of the current pixel, we search for pixels of the same class $\mathbf{x}_i = \{x_i^1, x_i^2, \dots, x_i^J\}$ as x_i by performing a non-local search (e.g., via block-matching [5]) of similar patches on the clean image. When the raw image data is arranged in a Bayer pattern (e.g., R-G-G-B), we first transform the raw image into four separate channels and then search for similar patches within each channel. Specifically, let the $\mathbf{X}_i$ denotes a local patch in the clean image which centers at x_i with size $p \times p$, and $\mathbf{P}_i$ denotes the nonlocal search window of size $q \times q$ within the clean image. As shown in Fig. 2, we search for J most similar patches $\mathbf{S}_i = \{\mathbf{X}_i^1, \mathbf{X}_i^2, \dots, \mathbf{X}_i^J\}$ for

$\mathbf{X}_i$ by computing the Euclidean distance within the search window $\mathbf{P}_i$, then locate the positions of center pixels $\mathbf{x}_i$ of these similar patches $\mathbf{S}_i$. In the associated noisy image, we can find the corresponding positions of $\mathbf{S}_i$ and obtain the paired noisy pixel y_i along with its similar pixels $\mathbf{y}_i = \{y_i^1, y_i^2, \dots, y_i^J\}$. Assuming the $\mathbf{v}$ as the pixel-wise variance map, then the variance v_i for noisy pixel y_i can be obtained by

$$v_i = \frac{\sum_{j=1}^J \|\mathbf{y}_i - \mathbf{x}_i\| - \text{Mean}(\mathbf{y}_i - \mathbf{x}_i)\|_2^2}{J}, \quad (3)$$

where $\text{Mean}(\cdot)$ denotes the averaging operation over set $\mathbf{S}_i$. The per-pixel variance estimation of the noisy image can be completed by moving the position of $\mathbf{x}_i$ through a sliding window across the whole image.

4.2. Variance estimation DCNN

As mentioned above, one of the most popular parametric models for real-world noise is the Poisson-Gaussian noise model in Eq. (1), in which the values of σ_r and σ_s are determined by the imaging settings (i.e., ISO, exposure parameters, camera models, as well as ambient conditions). From Eq. (1), one can see that the noise distribution is jointly determined by the intensity value of clean image pixels and the related imaging settings. Such nonlinear signal-dependent noise characteristics make the Poisson-Gaussian noise model often difficult to use in practical situations due to the following two reasons. On the one hand, the Poisson-Gaussian noise model is often not accurate enough to approximate the heterogeneous real-world noise. For example, due to the uncertainty factors involving photon counting, the signal-dependent noisy component possesses a nonlinear relationship with the observed noisy pixel as shown in Eq. (1). Untangling such dependencies is nontrivial, as previous work on unprocessing images [27] has demonstrated. Meanwhile, the Poisson-Gaussian model alone does not take other potential noise sources (e.g., quantization noise and banding pattern noise) into account. On the other hand, the problem of reliably estimating the parameters σ_r and σ_s in the Poisson-Gaussian model from a given noisy image is challenging, especially when they vary from one imaging setting to another.

In view of the limitations of the Poisson-Gaussian noise model, we propose to consider an alternative model for characterizing the heterogeneity of real-world noise by estimating the variance on a pixel-by-pixel basis. To demonstrate the effectiveness of the proposed noise modeling method, we use a simple DCNN architecture rather than a sophisticated neural network. Based on such observations, we propose a DCNN-based pixel-wise variance estimation network $\mathcal{F}(\cdot)$ to learn pixel-wise variance map $\tilde{\mathbf{v}}$ as following

$$\tilde{\mathbf{v}} = \mathcal{F}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\gamma}), \quad (4)$$

where $\mathbf{x}$ is the clean image, $\boldsymbol{\mu}$ and $\boldsymbol{\gamma}$ denote the ISO and camera label respectively.

Without any sophisticated design, we use a common 12-layer DCNN without any sophisticated design. In particular, the first layer uses 3×3 kernels to increase the number of feature channels from 6 to 64. Except for the last layer, other layers produce 64-channel feature maps with 3×3 kernels and ReLU nonlinearity. Moreover, the last layer uses 3×3 kernels to reduce the number of feature channels from 64 to 4 and outputs the estimated pixel-wise variance maps.

The architecture of the variance estimation network is as shown in Fig. 3. At the input, we feed the conditional signals (clean image, ISO, camera label) into the DCNN to learn the per-pixel variance map. Note that the proxy output (pixel-wise variance map) has been pre-computed as described in Section 4.1. To mark the image data acquired by different cameras, we assign them different numerical labels. Since the camera label and ISO are both scalar data, we represent the camera model label and ISO by a matrix of the same size as the clean image with repeated values.

4.3. Network training and extension

Note that we first compute the pixel-wise variance map by non-local means, and save the variance maps as labels of the variance estimation DCNN. We train the variance estimation DCNN on the training images of the proposed dataset by minimizing the following loss function,

$$L = \|\mathcal{F}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\gamma}) - \mathbf{v}\|_2^2, \quad (5)$$

where $\mathbf{v}$ denotes the pixel-wise variance map computed by non-local block-matching and $\mathcal{F}(\cdot)$ denotes the function of the variance estimation network.

In our current implementation, we set $J = 16$ in Section 4.1 which means we search for 16 most similar patches for the patch centered at a given clean pixel. The ADAM optimizer [46] was employed to train the network with parameter setting $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. We have set the initial learning rate as 10^{-4} and apply a strategy of learning rate decay. The hardware infrastructure employed in our study consists of high-performance computing resources, including a multi-core CPU and NVIDIA GeForce RTX 2080 GPU accelerators. Specifically, we utilized a dual-socket server equipped with Intel Xeon processors clocked at 2.2 GHz, providing a total of 48 CPU cores and 94 GB of RAM. In addition, the proposed network is implemented on the PyTorch platform and trained using one Nvidia GeForce RTX 2080 GPU.

5. Experiments

5.1. Experimental setup

The proposed variance estimation network was trained using the proposed dataset, which consists of 140 realistic clean-noisy image pairs. All collected pairs are divided into training and test parts, the training set contains 112 image pairs and the other 28 image pairs belong to the testing set, and the image pairs of the testing set contain all kinds of smartphones and ISO numbers. All images are cut into image patches with size 64×64 . To measure the level of similarity of simulated noise distribution and real noise distribution, we apply the metric of Kullback-Leibler (KL) divergence [18] between the artificial noise and real noise. To compare the denoising performance among synthetic data generated by different noise modeling methods, we use the testing set of the proposed dataset as our denoising test dataset. Moreover, we introduce the newly collected SIDD validation dataset released by the NTIRE 2020 Real Image Denoising Challenge [47] as the other denoising test set to demonstrate the generalization ability of denoising networks trained by different simulated noisy data, note that the SIDD validation dataset is not overlapped with the original SIDD dataset.

Furthermore, we conduct real-world noisy image classification experiments for the raw sensor data to verify the noise modeling ability of different methods. Specifically, the raw noisy training dataset originates from the ImageNet dataset [35]. Since the ImageNet dataset consists of sRGB images, we first extract the color filter array (CFA) data, and then add different synthetic noise to clean raw images to derive simulated noisy images. In this way, we could train several image classification networks using different kinds of simulated noisy data. To demonstrate the classification accuracy on realistic noisy images, we have built a real noisy image classification test set which contains 20 classes, and each class includes 50 images. The noisy images were captured by 3 different smartphones, namely Huawei P20, Redmi K40 and Apple iPhone 7 Plus.

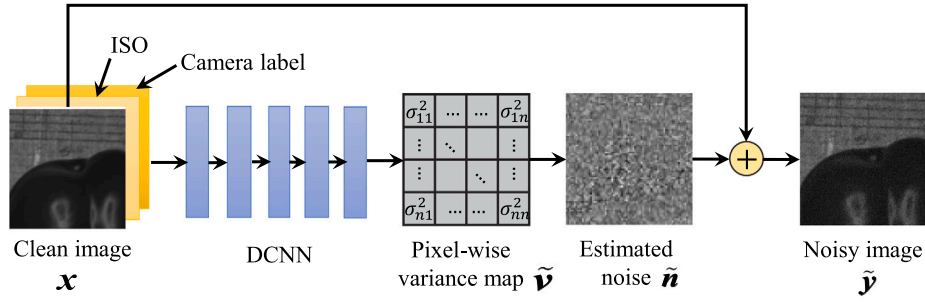


Fig. 3. The architecture of our proposed DCNN-based noise modeling method.

Table 3

Accuracy comparison of pre-computed variance maps.

Model	Poisson-Gaussian	Pixel-wise variance map
D_{KL}	0.0442	0.0159

Table 4

Effect of different network architectures.

$\tilde{n}$	2-layer CNN	Resnet	Unet	12-layer DCNN
D_{KL}	0.0256	0.0242	0.0231	0.0228
GFLOPs	0.02	1.53	1.47	1.54

5.2. Ablation studies

5.2.1. The accuracy of pre-computed pixel-wise variance map

The local pixel-wise variance maps are first computed by non-local means and serve as the proxy label of the variance estimation network. To make sure the pre-computed variance maps and non-i.i.d Gaussian noise assumption are plausible, we compute the KL-divergence between the synthetic noise generated by the pre-computed pixel-wise variance maps and the real noise, and compare it with the common Poisson-Gaussian noise model. The experimental results are shown in Table 3. According to the results of average KL-divergence, we can see that the noise generated by the pre-computed local variance map is much closer to the distribution of real-world noise than the Poisson-Gaussian noise model (D_{KL} decreases by a factor of three), which verifies the accuracy of our pixel-wise variance assumption.

5.2.2. The effect of different network architectures

To explore the influence of different network architectures on noise modeling ability, we have compared the average KL-divergence when employing different network architectures. Considering the simplest network, we first use a 2-layer CNN as the variance estimation network. In Table 4, one can see that the average KL-divergence is the largest when applying the 2-layer CNN. In order to explore the effect of network architectures exactly, we keep a similar computing consumption (GFLOPs) among other complicated networks, i.e., Resnet, Unet, and 12-layer DCNN. From the experimental results in Table 4, it could be observed that the variance estimation network could derive better noise modeling ability when applying the Unet and 12-layer DCNN. The small KL-divergence fluctuations among Resnet, Unet and 12-layer DCNN demonstrate that the noise modeling ability of the proposed method is not sensitive to the change of network architectures.

5.2.3. The effect of conditional signals

To guide the variance estimation network training, we need to feed conditional signals into the DCNN. To distinguish the necessary conditional signals, we have compared the performance when different conditional signals are fed into the DCNN to guide the training. Note that we consider the clean image necessary to predict the variance map from Eq. (1). From the results in Table 5, it can be observed that when

Table 5

Effect of conditional signals on the accuracy of noise modeling.

ISO	$\times$	$\checkmark$	$\times$	$\checkmark$
Camera label	$\times$	$\times$	$\checkmark$	$\checkmark$
D_{KL}	0.0719	0.0736	0.0654	0.0228

Table 6

The comparisons of average KL-divergence, number of parameters (M), and running times (s) of different noise modeling methods on the proposed test dataset.

Model	D_{KL}	No. of Para. (M)	Running time (s)
Gaussian	0.7275	/	/
Poisson-Gaussian	0.0442	/	/
Noise flow [18]	0.0505	0.0024	13.35
Ours	0.0228	0.38	8.58

the ISO and camera information are not provided, the noise estimated by the DCNN is not similar to the real noise by producing a large KL-divergence value. When either the ISO or camera information is given, the KL-divergence value could not greatly reduce, revealing that the guidance from the single ISO or camera information could not help the DCNN to improve the noise modeling ability effectively. The closest distribution to real-world noise is achieved when all conditional signals are fed into DCNN.

5.3. Comparison with state-of-the-art noise modeling methods

To verify the performance of the proposed noise modeling method, we first compare the KL-divergence between real-world noise and synthetic noise generated by different competing noise modeling methods. Our benchmark methods include both parametric noise models (i.e., Gaussian noise and Poisson-Gaussian noise [13]) and nonparametric noise model, i.e., Noise Flow [18]. Since the training code of the CA-NoiseGAN [16] method is unavailable, we could not make the performance comparisons with CA-NoiseGAN [16] temporarily. Regarding the Poisson-Gaussian model of the proposed dataset, we first estimate the hyperparameters of the Poisson-Gaussian model using the method of [13], then we can generate synthetic data of the Poisson-Gaussian model directly. We have re-trained the Noise Flow method using the same training data. The average KL-divergence comparisons between real-world noise distribution and different simulated noise distributions are shown in Table 6.

From the experimental comparisons of KL-divergence one can see that among the parametric noise models, the commonly used Gaussian noise distribution has the worst performance. By contrast, the smaller KL-divergence between the real-world noise and Poisson-Gaussian noise demonstrates that the Poisson-Gaussian noise model is more effective in modeling real-world noise by considering the signal-dependency of noise. Furthermore, the non-parametric Noise Flow method [18] obtained a slightly bigger KL-divergence result than the Poisson-Gaussian noise model due to the instability of the Noise Flow method. When compared to all competing methods, our method has achieved the

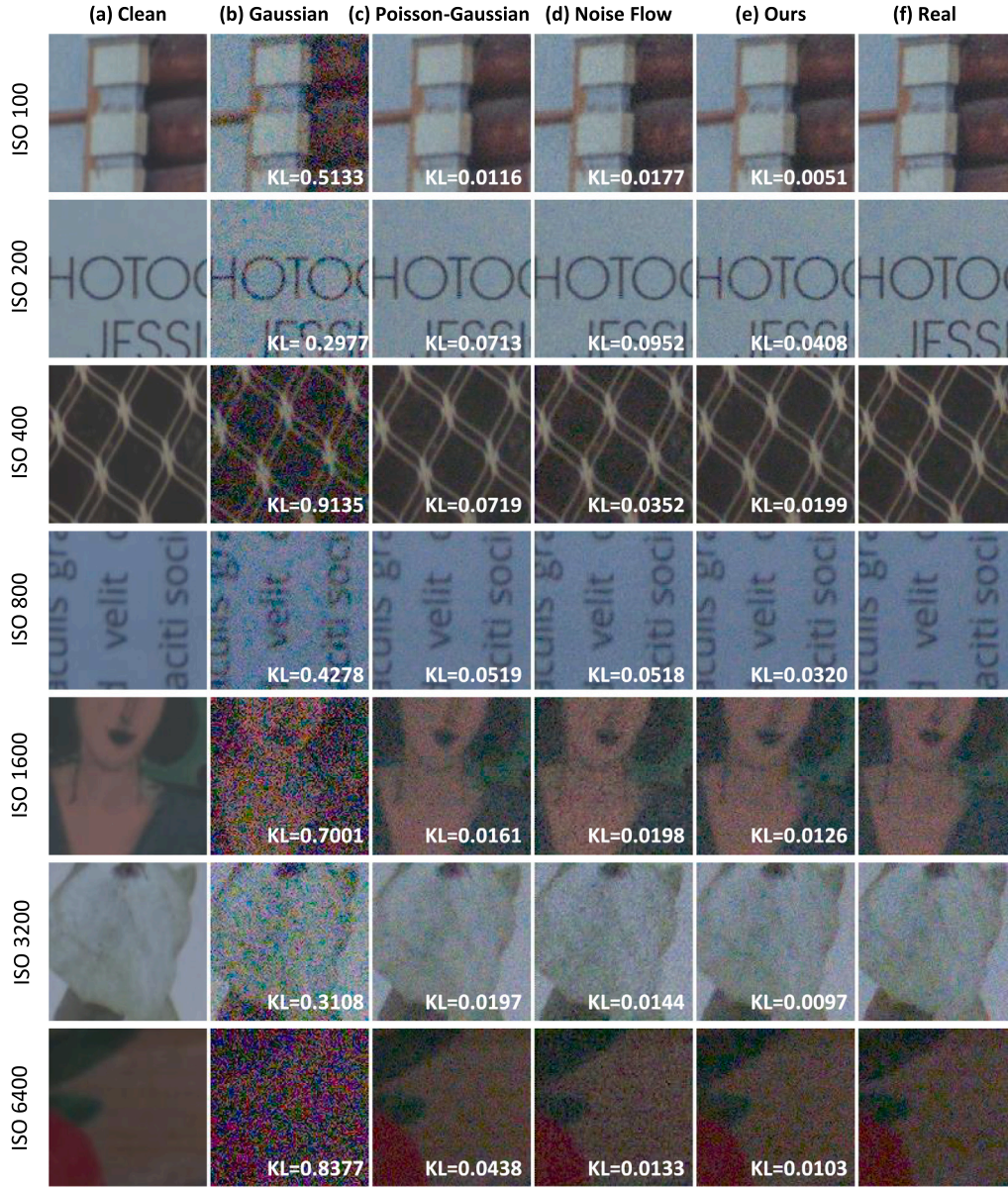


Fig. 4. Visual comparisons of noisy patches generated by: (b) Gaussian, (c) Poisson-Gaussian, (d) Noise Flow [18] and (e) Our method. The ISO parameters are shown on the left. These patches of raw-data passed through the ISP pipeline provided by SIDD [10] for visualization.

smallest KL-divergence, which verifies the effectiveness of the proposed method. The visual comparisons of simulated noisy images by various competing noise modeling methods are shown in Fig. 4; one can see that the noisy images generated by our method are more realistic.

5.4. Real-world image denoising

To further compare the performance of different noise modeling methods, we first generate artificial training data from different noise modeling methods and then train a common image denoising network with the same network structure (a similar strategy has been adopted in previous works [16,18]). By evaluating the denoising performance on real-world noisy images, we can more directly compare the accuracy of noise modeling with all competing methods. For the common architecture of the denoising network, we have adopted the DnCNN network [48] which contains 10 convolutional layers.

The denoising results are shown in Table 7. From the denoising results, we observe that DnCNN-Gaussian has the worst PSNR performance. Considering the results evaluated on the proposed test

Table 7

The real image denoising comparisons of DnCNNs trained with simulated data from different noise modeling methods on the proposed test dataset and SIDD validation dataset.

Model	Proposed test dataset		SIDD validation dataset	
	PSNR	SSIM	PSNR	SSIM
Noisy	40.73	0.9583	37.19	0.8361
Gauss	46.85	0.9735	41.79	0.8541
Poisson-Gaussian	49.64	0.9963	44.91	0.9267
Noise flow [18]	48.26	0.9942	43.75	0.9471
Ours	50.77	0.9973	47.72	0.9781
Real	50.67	0.9974	46.19	0.9673

dataset, the DnCNN trained by our method could achieve approaching denoising performance with the DnCNN trained by the real clean-noisy pairs, which verified the effectiveness of the proposed noise modeling method. On the other hand, the DnCNNs trained on the simulated noisy data were devoted to model the original sensor noise properties characterized by 6 specific smartphones, while the SIDD validation dataset

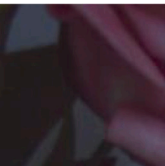
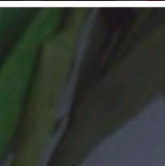
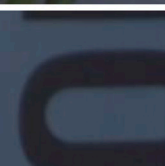
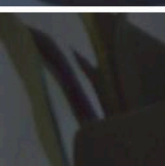
(a) Clean	(b) Gaussian	(c) Poisson-Gaussian	(d) Noise Flow	(e) Ours	(f) Real
	PSNR=49.47	PSNR=52.79	PSNR=51.61	PSNR=53.80	PSNR=53.64
	PSNR=36.93	PSNR=39.01	PSNR=37.38	PSNR=40.04	PSNR=39.12
	PSNR=43.92	PSNR=49.04	PSNR=47.04	PSNR=50.87	PSNR=50.96
	PSNR=46.46	PSNR=48.34	PSNR=46.58	PSNR=50.48	PSNR=50.85
	PSNR=46.42	PSNR=48.00	PSNR=46.90	PSNR=51.90	PSNR=51.83
	PSNR=44.36	PSNR=49.50	PSNR=47.10	PSNR=51.39	PSNR=50.96

Fig. 5. Visual comparisons of denoised patches recovered by different DnCNNs on the proposed test dataset. The DnCNNs were trained by noisy data from: (b) Gaussian, (c) Poisson-Gaussian, (d) Noise Flow [18], (e) Our method and (f) Real data. These patches of raw-data passed through the ISP pipeline provided by SIDD [10] for visualization.

was captured by 5 different smartphones [47]. From the experimental results on the SIDD validation dataset in Table 7, DnCNNs trained by different simulated noisy data could obtain various performance gains, revealing that the proposed dataset could characterize diverse noise properties. Moreover, the average PSNR gain of DnCNN-Ours over the DnCNN-Real is 1.53 dB, which demonstrates the proposed noise modeling method possesses a good generalization ability. Visual comparisons of the denoised images by different DnCNNs trained with simulated noisy data are shown in Figs. 5 and 6. From the visual comparisons of Fig. 5, the DnCNN-Ours can recover more image details than other DnCNNs trained by artificial noisy data sampled from competing noise modeling methods. Since the SIDD validation dataset represents different sensor noise characteristics from the proposed dataset, most DnCNNs could not remove the noise effectively as shown in Fig. 6, while DnCNN-Ours could recover a cleaner image revealing that the proposed method has a great generalization ability.

5.5. Real-world image classification

In addition to the real denoising task, we also have implemented experiments of the high-level task, i.e., image classification [49,50], to evaluate the accuracy of simulated noisy data generated by different

noise modeling methods. In particular, we generate several kinds of simulated noisy data using different noise modeling methods, then train respective noisy image classification networks for raw data with the same network architecture. Note that the training dataset comes from the ImageNet dataset [35], we first extract raw data from original images in ImageNet and add different artificial noise to clean images to obtain noisy raw images. To verify the real noisy image classification accuracy, we built a noisy classification test dataset of the real world, the details of the test dataset have been described in Section 5.1, and some visualizations of the captured test dataset have been shown in Fig. 7.

Recently, [51] proposed a joint image processing and classification pipeline for raw data, containing image denoising, demosaicking and classification. With domain-specific knowledge of the high-level task and end-to-end training, the method of [51] is proved to be more superior to traditional sequential methods. Inspired by [51], we employ a similar end-to-end denoising, demosaicking and classification network to explore the accuracy of different noise modeling methods. As shown in Fig. 8, the proposed network contains two subnets, one is used for image denoising and demosaicking, and the other is the image classification network. Since the input data is 4-channel noisy raw data, we first feed the noisy data into a regular denoising and demosaicking



Fig. 6. Visual comparisons of denoised patches recovered by different DnCNNs on the SIDD dataset. The DnCNNs were trained by noisy data from: (b) Gaussian, (c) Poisson-Gaussian, (d) Noise Flow [18], (e) Our method and (f) Real data. These patches of raw-data passed through the ISP pipeline provided by SIDD [10] for visualization.

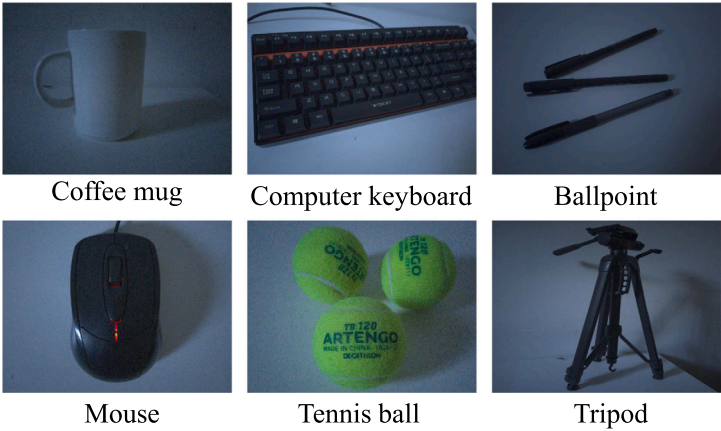


Fig. 7. Visualizations of realistic noisy images captured by smartphones from our classification test dataset. These raw-data images passed through the ISP pipeline provided by SIDD [10] for visualization (Please zoom in to see better details).

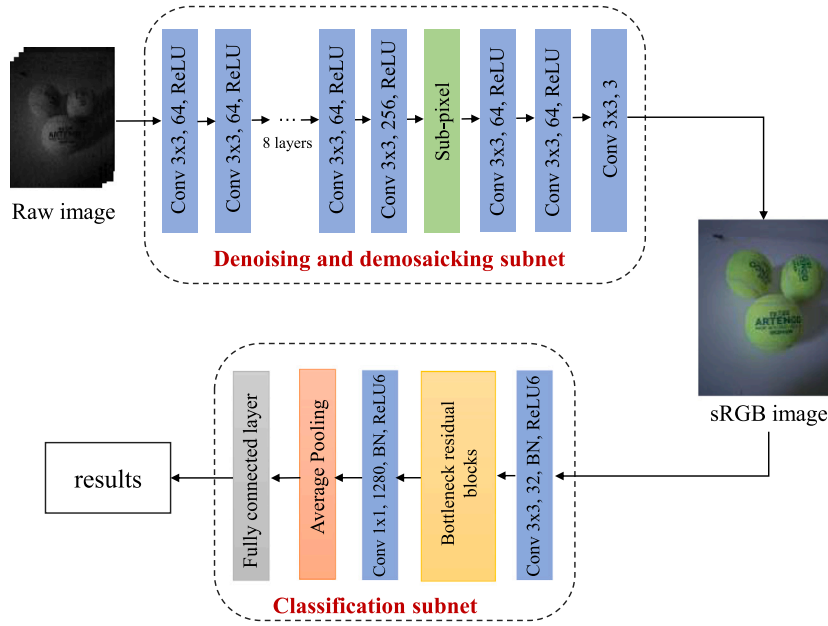


Fig. 8. The architecture of the overall real-world noisy image classification network.

Table 8

The real image classification comparisons of the joint demosaic and MobileNetV2 networks trained with simulated noisy data from different noise modeling methods on the proposed classification test dataset.

Model	Gaussian	Poisson-Gaussian	Noise flow [18]	Ours
Top 1	20.1%	24.0%	22.6%	25.2%
Top 5	31.5%	36.5%	38.7%	41.2%

network to estimate 3-channel sRGB data, and then feed the sRGB data into the classification network to output classification results. The architecture of the proposed network is demonstrated in Fig. 8, the denoising and demosaicking subnet has 15 convolutional layers, and each layer uses 3×3 kernels and follows a ReLU nonlinearity except the last layer, the first layer increases the number of feature maps from 4 to 64, the layers from 2th to 11th generate 64-channel feature maps, the 12th layer increase the number of feature maps from 64 to 256 to facilitate increasing the image resolution by the sub-pixel function [52], then 13th and 14th layers produce 64-channel feature maps, finally, the last layer uses 3×3 kernels to recover the 3-channel RGB image. Moreover, we use a lightweight classification network, i.e., MobileNetV2 [34], as the baseline classification network, where bottleneck residual blocks are essential components of MobileNetV2, more details of MobileNetV2 architecture can be found in [34].

The noisy image classification results are demonstrated in Table 8, one can see that the classification network trained by simulated noisy data generated by the proposed noise modeling method could derive the highest accuracy, revealing that our method can generate the most approaching noisy data with real noisy data. Note that the largest ISO number of common smartphones is 3200 or 6400, while digital cameras have a larger ISO range which could be up to 25 600. Therefore, the intensity of artificial noise is not so large as the noise captured by smartphones, and the performance gap among classification networks trained by different simulated noisy images is comparatively small.

5.6. Complexity analysis

We have compared the number of parameters and running time in Table 6. The running time only refers to the network inference time that samples noisy patches on a complete image for each competing non-parametric noise modeling method. From the comparisons in Table 6,

it could be observed that even though the Noise Flow method has a smaller number of parameters, the running time to generate noisy patches is still larger than our method, and our method achieves better performance.

5.7. Discussion

According to the above experimental results, the proposed noise modeling method can generate more realistic simulated noise compared to competing methods. The developed model works because we employ an efficient DCNN to estimate the pixel-wise variance map which effectively characterizes the heterogeneity of the real-world noise distribution at the finest pixel level, breaking through the limitations of parametric noise models which adjust parameters manually and effectively prompting the modeling accuracy. However, the non-i.i.d Gaussian distribution inevitably ignores some minor noise components that do not follow exactly the Gaussian distribution. Therefore, a more elaborate and accurate noise model is expected in future work.

6. Conclusion

In this paper, we propose a benchmark dataset of the real world and a novel noise modeling algorithm based on a variance map estimation network. Different from current real-world datasets, we consider more smartphone models and more shooting scenes that contain both indoor scenes and outdoor scenes. Good denoising generality ability on the SIDD validation dataset has demonstrated the diverse noise properties of the proposed dataset and the effectiveness of the proposed noise modeling method. In our noise modeling method, we train a variance estimation network capable of generating a variance map on a pixel-by-pixel basis. We have found that conditional signals (ISO and camera model) benefit the estimation of pixel-wise noise variance maps. Experimental results of KL-divergence, real-world denoising performance and real-world noisy image classification accuracy demonstrate the effectiveness of the proposed method. However, there is a weakness in our approach, i.e., we still need a few real-world clean and noisy image pairs under different noisy conditions to estimate the pixel-wise variance maps, yet the collection of real-world image pairs is generally time-consuming. Regarding the proposed method, the realization of real denoising consists of several separate processes, i.e., the computation of variance maps, the learning of the noise model and the training

of the real denoising model. In future work, a successive and efficient real denoising strategy is expected to be explored.

CRedit authorship contribution statement

Lu Sun: Methodology, Software, Validation, Writing – original draft. **Jie Lin:** Conceptualization, Formal analysis, Methodology, Supervision. **Weisheng Dong:** Conceptualization, Formal analysis, Writing – review & editing. **Xin Li:** Conceptualization, Supervision, Writing – review & editing. **Jinjian Wu:** Conceptualization, Supervision. **Guangming Shi:** Resources, Supervision.

Declaration of competing interest

The authors declared no potential conflicts of interest with respect to the research, author-ship, and publication of this article.

Data availability

The data that has been used is confidential.

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